

short position in the risk-free rate is 0.085. The corresponding certainty equivalent restricting the investor to only risky asset positions obtained earlier is 0.077. Letting the investor have access to the (short) risk-free asset position represents a significant risk-free utility increase of eighty basis points.

3.3. NON-PARTICIPATION IN THE STOCK MARKET

Mean-variance investing predicts that with just equities and a risk-free asset, all investors should invest in the stock market except those who are infinitely risk averse. In reality, only half of investors put money in the stock market.²² This is the *non-participation puzzle*.

Table 3.12 reports equity market participation rates by households in the United States calculated by Li (2009) using data from the Panel Study of Income Dynamics (PSID), a household survey conducted by the University of Michigan, and the Survey of Consumer Finances (SCF) conducted by the Federal Reserve Board. (Obviously you can't be poor if you have some savings, and researchers go further and exclude those with meager savings.)

In Table 3.12, the PSID and SCF stock holdings track each other closely and have hovered around 30%. The SCF also counts stocks included in pension plans and IRAs and when these retirement assets are included, the proportion of households holding stocks increases to around 50%. There has been a general increase in stock market participation when retirement assets are included from around 30% in the 1980s to 50% in 2005. Yet about half of U.S. households do not hold any equities. This is not just an American phenomenon; Laakso (2010) finds that stock market participation in Germany and France is well below 50% including both direct investments and those made indirectly through mutual funds and investment accounts. Italy, Greece, and Spain have stock participation rates at approximately 10% or less.

Several explanations have been proposed for the high non-participation in stocks markets. Among these are:

Table 3.12

Equity Market Participation Rates							
	1984	1989	1994	1999	2001	2003	2005
PSID	27%	31%	37%	28%	32%	29%	27%
SCF (excluding pensions and IRAs)	–	21%	23%	30%	32%	30%	29%
SCF (including pensions and IRAs)	–	32%	39%	50%	52%	51%	51%

²²The first paper in this literature is Blume, Crockett, and Friend (1974). The non-participation puzzle rose to economists' attention with Mankiw and Zeldes (1991) as an explanation for the equity premium puzzle, which I discuss in chapter 8.